Dr. Víctor Manuel Chávez Pérez

Professor–Researcher · Applied Physics, Data Science & Science Education

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Google Scholar · ResearchGate · LinkedIn

SNII · Candidate (SECIHTI) 2025–2029

PROFILE

I am a physicist from the University of Guadalajara and hold a PhD in Applied Physics from the University of Vigo. My research blends atmospheric and climate physics with large-scale data analysis: I study sudden stratospheric warmings (SSW), heat waves and their response under climate-change scenarios using models such as CMIP5 and WACCM.

Beyond basic research, I bring physics to real-world problems: I am developing a flood early-warning system for the Guadalajara Metropolitan Area that integrates weather radar, hydrological runoff modeling over high-resolution LiDAR terrain and remote sensing. My work also spans physical oceanography, remote sensing and applied optics.

I am also interested in how generative AI can transform science education: I design learning assistants and teaching materials that I evaluate with students themselves. Alongside this, I co-founded the Tlakakini Science & Technology Club, devoted to science outreach for children. I believe the best science is the kind that gets shared.

RESEARCH INTERESTS

Sudden stratospheric warmings · Stratospheric dynamics · General circulation models (CMIP5 / WACCM) · Climate change · Physical oceanography & remote sensing · Applied optics · Data science · High-performance computing · Science education & outreach · Generative AI in science education · Machine Learning applied to climate & atmospheric data · Student perception of AI-based learning tools

EDUCATION

2017 – 2024 **PhD in Applied Physics**
Universidade de Vigo · Spain
Thesis: Sensitivity of sudden stratospheric warmings to simulation and analysis conditions in models.

2011 – 2012 **MSc in Climate Science: Meteorology, Physical Oceanography & Climate Change**
Universidade de Vigo · Spain
Temporal and spatial variability of the Minho river plume (MODIS 250 m). Top honors.

2009 – 2011 **MSc in Hydrometeorology — Meteorology & Physical Oceanography**
Universidad de Guadalajara · Mexico
Seasonality and anomalies of oceanographic variables in the Gulf of Mexico via remote sensing.

2000 – 2008 **BSc in Physics**
Universidad de Guadalajara · Mexico
Thesis: Ocean and coastal meteorology of Nayarit (San Blas, 1999–2002). Grade 95/100.

EXPERIENCE

2026 – present **Adjunct Professor ("Profesor de Cátedra")**
Tecnológico de Monterrey · Campus Guadalajara · Zapopan, Jalisco · Mexico
Course unit F1025B, The Laws of Motion and Conservation, team-taught by two instructors — one for theory and one for the challenge. Two theory groups and one challenge group.
Courses: The Laws of Motion and Conservation (Theory) ×2 · The Laws of Motion and Conservation (Challenge)

2024 – present **Lecturer "B"**
Centro Universitario de Tlaquepaque · UdeG · Tlaquepaque, Mexico
Department of Basic Sciences. Chair of the Basic Sciences Academic Board.
Courses: Calculus ×2 · Advanced Calculus ×2 · Numerical Methods ×2 · Probability · Electromagnetism Lab · Discrete Mathematics

2024 – present **External Collaborating Researcher**
Universidade de Vigo · EPhysLab · Ourense, Spain
Doctoral research on sudden stratospheric warmings in CMIP5 and WACCM models; statistical analysis of large climate datasets.

2025 – 2026 — **Lecturer**
Centro Universitario Bonpland y Humboldt · Online
PhD in Aerospace Engineering: Research Methodology and Fluid Mechanics.
Courses: Research Methodology ×2 · Fluid Mechanics

2019 – 2026 — **Adjunct Professor ("Profesor de Asignatura B")**
Universidad Tecnológica de Jalisco · Guadalajara, Mexico
Mechatronics Division — Basic Sciences.
Courses: Engineering Physics ×10 · Probability & Statistics ×8 · Engineering Mathematics II ×6 · Differential Calculus ×5 · Integral Calculus ×3 · Engineering Mathematics I ×2 · Multivariable Calculus · Applied Engineering Statistics · Thermodynamics

2019 — **Faculty Lecturer**
Tecmilenio · Campus Guadalajara · Guadalajara, Mexico
High school: Matter & Energy, Integral Calculus, Science & Technology.
Courses: Integral Calculus ×2 · Differential Calculus · Mathematics (Language of Science) · Matter & Energy · Science & Technology

2019 — **Physics Instructor**
UNITEC · Campus Guadalajara · Guadalajara, Mexico
Engineering programs: Statics.
Courses: Statics

2015 — **Predoctoral mobility stay**
Universidad Pablo de Olavide · Seville, Spain
Short research stay at R&D centers; results presented at international conferences.

2013 – 2017 — **Predoctoral Researcher (FPI Fellowship)**
Universidade de Vigo · EPhysLab · Ourense, Spain
Statistical analysis of large-volume data for climate-change research with general circulation models.

2013 – 2014 — **Predoctoral research stay**
Universidad de Guadalajara · Guadalajara, Mexico
Academic research stay on climate-change analysis.

2010 – 2012 — **Physics & Math Instructor**
Centro de Enseñanza Técnica Industrial (CETI Colomos) · Guadalajara, Mexico
Engineering programs.
Courses: Differential & Integral Calculus ×2 · Statics ×2 · Dynamics ×2 · Mathematics ×2

2010 – 2011 — **Physics & Math Instructor**
Universidad de Especialidades · Guadalajara, Mexico
Mechanics, Introduction to Physics and Electromagnetism.
Courses: Mechanics · Intro to Physics · Electromagnetism

2008 – 2011 — **Professor–Researcher**
Instituto Tecnológico Superior de Zapopan · Zapopan, Mexico
Mathematics for engineering; parallel computing applied to oceanography.
Courses: Physics (Mechanics, E&M, Thermodynamics) ×11 · Mathematics & Calculus (Differential, Integral, Vector, ODEs) ×17 · Dynamics

2006 – 2007 — **Research Assistant**
Universidad de Guadalajara · CUCEI, Departamento de Física · Guadalajara, Jalisco · Mexico
"Oceanografía de la Costa de Nayarit" project (CONACYT-SEP-2003-C02-44870F). PAY and PITAI programmes.

2005 – 2006 — **Physics, Chemistry and Mathematics Teacher**
Centro Universitario UTEG · Guadalajara, Jalisco · Mexico
Intensive semi-schooled high school programme (BIS).

PUBLICATIONS

2025 — **Observación Astronómica en México: Pasando de lo Visible a lo Invisible**
Almaguer-Gómez C., Medina J. F., Chávez-Pérez V. M., et al. · *Óptica Pura y Aplicada 58(1):15* · DOI: 10.7149/OPA.58.1.51200

2025 — **Influence of Satellite and Presatellite Periods on the Validation of Major Sudden Stratospheric Warmings in Historical CMIP5 Simulations**
Chávez-Pérez V. M., Añel J. A., et al. · *Atmosphere 16(5)* · DOI: 10.3390/atmos16050628

2025 — **Temporal Variability of Major Stratospheric Sudden Warmings in CMIP5 Climate Change Scenarios**
Chávez-Pérez V. M., et al. · *Climate 13(10)* · DOI: 10.3390/cli13100207

2023 **Estacionalidad y Anomalías del Nivel del Mar en el Caribe por Medio de Datos de Altimetría**
Palacios Hernández E., Chávez Pérez V. M., Sierra Romero L. L. · *Generis Publishing* · ISBN: 979-8-88676-642-4

2022 **Impact of Increased Vertical Resolution in WACCM on the Climatology of Major Sudden Stratospheric Warmings**
Chávez-Pérez V. M., Añel J. A., et al. · *Atmosphere 13(4)* · DOI: 10.3390/atmos13040546

2018 **The climatology of dust events over the European continent using data of the BSC-DREAM8b model**
Mandija F., et al. · *Atmospheric Research 209:144–162* · DOI: 10.1016/j.atmosres.2018.03.006

In preparation

2026 Science Beyond the Classroom: Impact Assessment of a Monthly Science Club During National Non-School Days in Mexico
Almaguer-Gómez C., Chávez-Pérez V. M., et al.

2026 Student Perception of a Generative AI-Based Calculus Learning Assistant: A Mixed-Methods Study
Chávez-Pérez V. M., et al.

2026 Assessment of the lightning activity response to aerosol optical depth
Mandija F., Chávez-Pérez V. M.

CONFERENCES & TALKS

2022 **65th Mexican National Physics Congress**
2–7 oct 2022 · Zacatecas, Mexico
• Analysis of the occurrence frequency of Southern Hemisphere stratospheric sudden warmings in high-top CMIP5 models
• Optical characterisation of transparent 3D-printing resin

2020 **63rd Mexican National Physics Congress**
4–9 oct 2020 · Online
• Significance of basic SSW characteristics when comparing CMIP5 models with reanalysis data over pre-satellite and post-satellite periods
• Impact on the occurrence frequency of major stratospheric sudden warmings under different climate change scenarios using CMIP5

2018 **American Geophysical Union (AGU) Fall Meeting 2018**
10–14 dic 2018 · Washington, USA
• Impact of Increased Vertical Resolution in WACCM on the Climatology of Major Stratospheric Sudden Warmings

2018 **Mexican Geophysical Union Annual Meeting**
28 oct–2 nov 2018 · Puerto Vallarta, Mexico
• Changes in the climatology of major stratospheric sudden warmings using WACCM with high vertical resolution
• Assessment of the ability to reproduce major stratospheric sudden warmings in the CMIP5 historical dataset
• Major stratospheric sudden warmings under climate change scenarios with CMIP5

2017 **European Geosciences Union (EGU) General Assembly 2017**
23–28 abr 2017 · Vienna, Austria
• Estimation of the variability of dust aerosol optical depth over several European regions based on forecast model data

2016 **59th Mexican National Physics Congress (SMF)**
2–7 oct 2016 · León, Mexico
• Stratospheric teleconnection and stratospheric sudden warmings
• Characteristic patterns in the stratospheric variables of the Quasi-Biennial Oscillation
• Stratospheric anomalies before and after stratospheric sudden warmings for the different phases of the Quasi-Biennial Oscillation

2016 **SPARC Workshop SHARP 2016 — Stratospheric Change and its Role for Climate Prediction**
11–15 jul 2016 · Berlin, Germany
• Changes in the stratosphere before and after SSW events in the different phases of the Equatorial QBO
• Differences in the detection and classification of the Stratospheric Sudden Warming over the three reanalyses for the period 1979–2014

2015 **European Geosciences Union (EGU) General Assembly**
12–17 abr 2015 · Vienna, Austria
• Variability of the Brewer–Dobson circulation during Stratospheric Sudden Warmings

2015 **SPARC Regional Workshop — Role of the stratosphere in climate variability and prediction**
12–13 ene 2015 · Granada, Spain
• Variability of the Brewer–Dobson circulation during Stratospheric Sudden Warmings

2010 **Mexican Geophysical Union Annual Meeting**
7–12 nov 2010 · Puerto Vallarta, Mexico
• Seasonality and anomalies of oceanographic remote-sensing variables in the Gulf of Mexico
• Seasonality and anomalies of sea surface height in the Caribbean Sea using altimetry data
• An alternative for computationally intensive oceanographic data processing through parallel programming in MATLAB

2010 **53rd Mexican National Physics Congress**
25–29 oct 2010 · Boca del Río, Veracruz, Mexico
• High-performance computing applied to parallel programming to optimise processes in Astrophysics and Physical Oceanography

2009 **52nd Mexican National Physics Congress (SMF)**
26–30 oct 2009 · Acapulco, Guerrero, Mexico
• Frequency spectra of meteorological data from María Madre Island
• Comparison of coastal and marine winds in the coastal zone of San Blas, Nayarit (1999–2002)

ACTIVE RESEARCH PROJECTS

2026 – present **Flood early-warning system for the Guadalajara Metropolitan Area — Principal investigator**
Centro Universitario de Tlaquepaque de la Universidad de Guadalajara · Funding: No external funding
Development of an urban flood early-warning system that combines remote sensing (IAM-UdeG weather radar), hydrological runoff modeling over high-resolution LiDAR terrain, and official flood inventories. The system classifies risk by sub-basin in real time and estimates water arrival time, aiming to calibrate thresholds through a retrospective analysis with historical Civil Protection/Fire Department records.

2026 – present **Generative AI for lab practice reports (Google Gems) — Principal investigator**
Centro Universitario de Tlaquepaque de la Universidad de Guadalajara · Funding: No external funding
Implementation of Google Gems as a writing and feedback assistant for students to draft and self-assess their lab practice reports in physics courses.

2025 – present **AI-generated educational comics for science teaching — Principal investigator**
Universidad de Guadalajara · Funding: No external funding
Development and evaluation of educational comics created with generative AI tools as a didactic resource for teaching physics, mathematics and science at basic and upper-secondary levels.

2025 – present **Pedagogical Calculus Assistant with Google Gems: development and student perception — Principal investigator**
Universidad Tecnológica de Jalisco · Funding: Universidad Tecnológica de Jalisco
Design and implementation of a Calculus learning assistant based on Google Gems, with quantitative and qualitative study of students' perception, satisfaction and academic performance.

2025 – 2028 **Optical elements to enhance visual acuity via wavefront coding — Collaborator**
Universidad de Guadalajara · Funding: Universidad de Guadalajara
Design and validation of optical elements that improve visual acuity through wavefront coding.

2025 – present **World Wide Lightning Location Network (WWLLN) — Collaborator · antenna lead**
University of Washington · Funding: University of Washington
Global lightning location network. In charge of the receiving antenna/node and real-time data delivery.

2024 – present **Tlakakini Science & Technology Club — Coordinator**
CUCEI · Universidad de Guadalajara · Funding: University of Guadalajara · CUCEI
Science-outreach club for children, with monthly experiment and workshop sessions.

TECH DEVELOPMENTS

Flood early-warning system · Guadalajara Metro Area
Python · FastAPI · Leaflet · NumPy · pysheds · rasterio · GDAL · Docker
Real-time web platform that anticipates urban flooding in the Guadalajara Metropolitan Area. It fuses the IAM-UdeG weather radar (one scan per minute) with a hydrological runoff model built on 5 m LiDAR and Copernicus DEM terrain, plus the IMEPLAN official flood inventory (363 recurrent sites). It produces a per-zone risk light —calibrated against official water-depth records— and a continuous risk grid that routes rainfall downslope, with concentration time and a "water incoming" ETA. Includes live and historical-replay views. Deployed in production.

COMMUNITY BUILDING & SERVICE

Thesis supervision

2011 **Estacionalidad y anomalías del nivel del mar en el Caribe por medio de datos de altimetría**
Lilia Lizeth Sierra Romero · Bachelor's · Co-advisor · Universidad de Guadalajara

Journal peer review

2025 **IEEE Latin America Transactions**
Reviewer — Q2 journal, double-blind

2024–2025 **Int. J. of Environment and Pollution (Inderscience)**
Reviewer — Q4 journal, double-blind

2025 **Congreso de Ingeniería, Ciencia y Gestión Ambiental (AMICA)**
Scientific committee reviewer

Project evaluation

2025–2026 **SECIHTI**
External evaluator — 2026 National Call, Basic & Frontier Science (3 proposals)

SCIENCE OUTREACH

2024 – **Tlakakini Science & Technology Club**
Co-founder. Monthly outreach sessions for children on national non-school days.

2009 – **Physics conferences (AGU, EGU, SMF)**
Talks at national and international conferences on climate and the stratosphere.

2005 – **Training science communicators**
Optics and electromagnetism workshops; National Outreach Meeting.

2009 – **Red Temática de Medio Ambiente y Sustentabilidad (Environment & Sustainability Thematic Network)**
Member of the CONACYT environment and sustainability network.

2011 – **Environmental Physics Laboratory (EPhysLab)**
External collaborating researcher with the University of Vigo group.

GRANTS & HONORS

2025–2029 **National Researcher System · Candidate**
SECIHTI · Mexico

2012–2016 **FPI Grant · Predoctoral Researcher**
Ministerio de Educación · Spain

2011–2012 **Climate Science Master's Scholarship**
Campus do Mar · Spain

2011–2012 **«Ourense Exterior» mobility grant**
Diputación de Ourense · Spain

2009–2011 **Hydrometeorology Master's Scholarship**
CONACYT · Mexico

2009 **Admission to the Sistema Estatal de Investigación (Jalisco State Research System)**
COECYTJAL · Jalisco, Mexico

2009 **Bachelor's Thesis Scholarship**
SEP · Mexico

CONTINUING EDUCATION

2025 **Diploma** — Techno-pedagogical strategies for developing 21st-century skills · Universidad de Guadalajara

2025 **Course** — Artificial Intelligence for teaching · Universidad de Guadalajara

2025 **Course** — Generative Artificial Intelligence for teachers · Universidad Anáhuac

2025 **Course** — Generative AI in the classroom · UNAM

2025 **Course** — Python for Data Science, AI & Development · IBM

2025 **Certification** — What is Data Science? · IBM

2025 **Course** — Differential Calculus for Data Science and AI · Platzi

2025 **Course** — Statistics and Probability · Platzi

2025 **Course** — Foundations of Software Engineering · Platzi

2025 **Course** — Introduction to the Terminal and Command Line · Platzi

2014 **Course** — OOP in Atmospheric and Oceanic Model Development · Universidade de Vigo

2011 **Diploma** — Teaching Competencies: Introduction to General High School Education · Universidad de Guadalajara

2005 **Workshop** — Training for Science Communicators · Universidad de Guadalajara

2025 **Course** — Mathematical Functions for Data Science and AI · Platzi

2005 **Course** — Theoretical and Conceptual Development of Electromagnetism (Part I) · Universidad de Guadalajara

2009 **Training** — DOCA Module I: The National System of Technological Education · Universidad de Guadalajara

LANGUAGES

Spanish — Native · English — Intermediate

TOOLBOX

Python · Fortran · MatLab · IDL · NCL · LaTeX · COMSOL · LabVIEW · MPI · Linux · Moodle · Pandas · Scikit-learn · Jupyter